

PlayIT: Offline-First Gamified Early Literacy Mobile Application

MVP Validation Framework & Instrument Specification (UPA Model — Weeks 1–2 Deliverable)

Course: IT411 — Capstone & Research 2 | Semester 1, AY 2026–2027
Institution: College of Computer Studies, Cebu Institute of Technology – University
Primary Framework: UPA (Usability, Pedagogy, Accessibility) | Standard 10-Item SUS + Child Smileometer + Pre/Post Phonics Test

1. Executive Summary & Research Rationale

Early childhood literacy education in Philippine public schools faces persistent challenges, including high pupil-to-teacher ratios, inadequate learning materials, and the digital divide characterized by limited or unstable home internet connectivity. While the Department of Education (DepEd) emphasizes the **Marungko Approach**—a phono-syllabic reading technique that introduces letters based on phonemic frequency rather than alphabetical order—many learners lack access to interactive, individualized phonetic practice at home. **PlayIT** addresses this gap as a 100% offline-first Android mobile application. It pairs the pedagogical structure of the Marungko Approach with modern offline speech recognition (Vosk), engaging gamification, and diagnostic parental telemetry.

The primary objective of the **Weeks 1–2 MVP Validation** is **formative and diagnostic**: (1) evaluate pediatric usability and child task flow; (2) quantify empirical reading gains through structured pre/post assessments; (3) secure expert validation from DepEd reading specialists regarding the 26-letter English Marungko adaptation; and (4) verify offline software stability across target mobile devices before feature lock.

2. Theoretical Evaluation Framework: The UPA Model

In educational technology for young learners, generic UI heuristics alone are insufficient. PlayIT adopts the **UPA Framework (Usability, Pedagogy, Accessibility)** as its primary overarching evaluation architecture across four distinct stakeholder cohorts:

- **Usability (U):** Evaluates navigation ease, interaction efficiency, and user satisfaction. Operationalized via the **Child Smileometer** (3-point visual scale — Read & MacFarlane, 2006) for young learners, the standardized **10-Item System Usability Scale (SUS — Brooke, 1996)** for parents and teachers, and observational task timings (time-on-task, vocal latency).
- **Pedagogy (P):** Evaluates curriculum alignment, phonetic accuracy, and reading efficacy. Operationalized via the **DepEd Teacher Pedagogy Checklist** (7 Likert items validating Marungko sequence, phoneme modeling, and CVC decodability) and the **Child Pre/Post Phonics Assessment** measuring empirical letter sound and word-blending score gains.
- **Accessibility (A):** Evaluates physical ergonomics and socioeconomic equity. Operationalized via **Pediatric Touch-Target Standards** (>=64dp buttons for motor control), high-contrast visual cues (WCAG 2.1 AA), audio-visual dual-coding, and **100% Offline Zero-Data Operability** with zero ads or tracking.

3. Curriculum Scope: 26-Letter Marungko English Adaptation & 33 CVC Matrix

PlayIT adapts the 28-letter Filipino Marungko sequence to **26 letters (A–Z)** for Grade 1 English phonics. The Spanish-derived grapheme **Ñ** does not exist in English. The digraph **NG (/ŋ/)** is a phonogram rather than a single letter; introducing it as an isolated grapheme creates phonological confusion for beginner readers. The 26 letters are organized into **7 progressive chapters** with **33 decodable CVC Blend It words**:

Chapter	Target Letters	Count	Biome Theme	Canonical Blend It Words (CVC Validated & Seeded in DB)
Chapter 1	m, s, a, i	4	Guava Greenery	SAM, SIS, AIM (3 words — Group 1 accepted exception)
Chapter 2	o, b, e, u	4	Mango Orchard	BUS, SUB, MOM, BEE, BIB
Chapter 3	t, k, l, y	4	Chocolate Hills	BAT, MAT, KIT, TOY, BOY
Chapter 4	n, g, p	3	Palm Valley	PIG, PAN, BUG, PIN, NAP
Chapter 5	r, d, h, w	4	Rice Terraces	DOG, HAT, HEN, BED, WEB
Chapter 6	c, f, j	3	Coral Reef	CAT, FAN, CAP, CUP, JAM
Chapter 7	q, v, x, z	4	Mount Pulag Summit	VAN, BOX, FOX, ZOO, QUIZ
Total	26 Letters	26	7 Biomes	33 Decodable CVC Target Words (100% Seeded in Room DB v3)

4. SMART Evaluation Objectives & Measurable Metrics

Dimension	Project Objective	Evaluation Metric & Instrument	Target Threshold	Cohort
Specific (S)	Verify child learning & task completion across core sublevels.	Task completion rate; 1st-pass ASR vocal accuracy; time-on-task.	>=85% task completion without help; >=75% ASR first-pass accuracy.	Early Learners (N=12)
Measurable (M)	Quantify usability & caregiver acceptance with standard scales.	10-Item SUS score; Child Smileometer; TAM constructs (PU, BI).	Mean SUS >=75.0 (Grade B+); Smileometer >=2.6/3.0; TAM PU >=4.2/5.0.	Parents & Teachers (N=15)
Achievable (A)	Validate curriculum alignment & measure empirical learning gain.	Teacher Pedagogy Checklist (7 items); Child Pre/Post Phonics Assessment.	100% agreement (>=4.0/5.0); Statistically significant gain (p < 0.05).	DepEd Teachers (N=5) + Learners
Relevant (R)	Ensure 100% offline persistence, data privacy, and diagnostic speed.	Room DB persistence audit; PDF export time; Noise gate test.	0% data loss on force-close; PDF export <=3.0s; Noise alert at >40dB.	Technical Evaluators (N=3)
Time-Bound (T)	Execute validation trials, analyze data, and refactor SRS/SDD.	Milestone schedule mapped to IT411 Capstone 2 calendar.	Protocol: Sept 13; Field Trials: Sept 18; Refactored SRS/SDD: Sept 19.	Research Team (Total N=30)

5. Stakeholder Mapping & Questionnaire Item Cross-Walk (UPA Model)

Code	Evaluation Criterion / Question Item	UPA Construct	Target Respondent	Scale
PED-01	Letter progression strictly follows DepEd Marungko frequency approach.	Pedagogy: Sequence	Teachers / SMEs	1–5 Likert
PED-02	Hear It delivers clear, natural phoneme sound modeling.	Pedagogy: Audio Model	Teachers / SMEs	1–5 Likert
PED-03	Say It uses age-appropriate example words (Mouse, Sun).	Pedagogy: Vocalization	Teachers / SMEs	1–5 Likert
PED-04	Find It uses culturally familiar, unambiguous illustrations.	Pedagogy: Visual Discrim.	Teachers / SMEs	1–5 Likert
PED-05	Blend It uses decodable CVC words respecting cumulative letters.	Pedagogy: Blending	Teachers / SMEs	1–5 Likert
PED-06	Excluding NG and Ñ for 26-letter English phonics is pedagogically sound.	Pedagogy: Scope Validity	Teachers / SMEs	1–5 Likert
PED-07	Non-punitive gamification (hearts/stars) reinforces mastery resilience.	Pedagogy: Motivation	Teachers / SMEs	1–5 Likert
PRE-POST	Child phoneme identification and CVC blending diagnostic assessment.	Pedagogy: Learning Gain	Early Learners (Tested)	0–10 Score Delta
SUS 1–10	Standard 10-Item System Usability Scale (Brooke, 1996).	Usability: Standard SUS	Parents & Teachers	1–5 Likert (0–100)
SMILEY	Child Smileometer affective rating (Sad=1, Neutral=2, Happy=3).	Usability: Affective	Early Learners	3-pt Visual Scale
OBS 1–5	Observational task completion & time-on-task (Map, Hear, Say, Find, Blend).	Usability: Efficiency	Child (Observed)	3-pt + Seconds
ACC-01	Pediatric touch targets (>=64dp) prevent miss-taps for small hands.	Accessibility: Ergonomics	Parents / Teachers	1–5 Likert
ACC-02	Audio-visual dual-coding enables pre-readers to navigate independently.	Accessibility: Dual-Coding	Parents / Teachers	1–5 Likert
ACC-03	High-contrast UI colors and large text meet WCAG 2.1 AA standards.	Accessibility: Contrast	Parents / Teachers	1–5 Likert
ACC-04	100% offline access ensures equal opportunity with zero mobile data cost.	Accessibility: Equity	Parents / Guardians	1–5 Likert
ACC-05	Ad-free, zero in-app purchase environment guarantees child safety & privacy.	Accessibility: Safety	Parents / Guardians	1–5 Likert
SYS 1–3	Technical verification: ASR latency <=0.5s, 60 FPS UI, Room DB integrity.	Technical: ISO 25010	Technical Evaluators	Pass / Fail / ms

6. Detailed Operationalized Instruments (UPA Model)

Part A: Pedagogy (P) — DepEd Teacher Checklist & Child Pre/Post Test

1. DepEd Teacher & SME Phonics Checklist (5-Point Likert: 1=Strongly Disagree to 5=Strongly Agree):

- **[PED-01] Sequential Phonics Progression:** The letter progression strictly follows the sequential DepEd Marungko frequency-based approach.
- **[PED-02] Phonemic Audio Quality:** Hear It delivers accurate, natural, and clear phoneme sound modeling suitable for beginner Filipino readers.
- **[PED-03] Vocal Production Practice:** Say It uses age-appropriate example words (e.g., Mouse, Sun) that encourage verbal phonics production without frustration.
- **[PED-04] Visual Picture Discrimination:** Find It uses illustrations that are culturally familiar, unambiguous, and easily recognizable to Filipino children.
- **[PED-05] Word Blending Feasibility:** Blend It uses decodable CVC words (e.g., MOM, KIT, PIG, HEN, CAP, WEB) respecting cumulative letter availability.
- **[PED-06] Curriculum Scope Validity:** Excluding NG (digraph) and Ñ (Spanish loan) to focus on 26 English letters is pedagogically sound for Grade 1.
- **[PED-07] Non-Punitive Gamification:** Hearts, stars, and gentle retry states support positive learning resilience without cognitive overload.

2. Child Empirical Pre/Post Phonics Diagnostic Assessment Protocol:

Administered before the child interacts with PlayIT (Pre-Test) and immediately following gameplay (Post-Test):

Sub-Test	Diagnostic Stimulus Items	Instruction & Administration Protocol	Scoring Rubric
Phoneme Sound Recognition	Letter flashcards: M, S, A, I, B	Show card: "What sound does this letter make?" (Prompt /m/, /s/)	1 pt per correct sound (Max: 5 pts)
Phonetic Picture Match	Picture cards: Mouse, Sun, Apple	Show 3 images: "Which picture starts with the /m/ sound?"	1 pt per correct match (Max: 3 pts)
CVC Word Blending	Word cards: SAM, BUS	Show word: "Can you sound out and read this word out loud?"	1 pt per blended word (Max: 2 pts)
Total Diagnostic Score	10 Total Phonics Items	Calculate empirical score gain: $\Delta = \text{Post-Test Score} - \text{Pre-Test Score}$	Total Score: 0 to 10 Points

Part B: Usability (U) — Standard 10-Item SUS & Child Smileometer

1. **Child Smileometer (Read & MacFarlane, 2006):** Show the 3 mascot emotion cards immediately post-session: "How did you feel playing with Lily today?" [1 = Sad/Difficult, 2 = Neutral/Okay, 3 = Happy/Fun]. Target: Mean Score $\geq 2.6 / 3.0$ (Positive Affective Response).

2. **Parent & Teacher System Usability Scale (Brooke, 1996):** Standard 10 items (1=Strongly Disagree to 5=Strongly Agree):

- **[SUS-01]** I think that I would like to use PlayIT frequently for my child's/students' reading practice.
- **[SUS-02]** I found the application unnecessarily complex to understand or operate. [Reverse-Scored]
- **[SUS-03]** I thought the application was easy to use and navigate.
- **[SUS-04]** I think that I would need the support of a technical person to be able to use this app. [Reverse-Scored]
- **[SUS-05]** I found the various functions in this system (Hear, Say, Find, Blend, Dashboard) were well integrated.
- **[SUS-06]** I thought there was too much inconsistency in the interface, audio, or navigation. [Reverse-Scored]
- **[SUS-07]** I would imagine that most children and parents would learn to use this system very quickly.
- **[SUS-08]** I found the application very cumbersome or awkward to use. [Reverse-Scored]
- **[SUS-09]** I felt very confident using the application and guiding my child through lessons.
- **[SUS-10]** I needed to learn a lot of things before I could get going with this application. [Reverse-Scored]

*Scoring Formula: For positive items (1,3,5,7,9), $\text{Score} = \text{Scale} - 1$. For reverse items (2,4,6,8,10), $\text{Score} = 5 - \text{Scale}$. Composite SUS = (Sum of Scores) x 2.5 (Normalized 0–100 scale; Target: Mean $\geq 75.0 / \text{Grade B+}$).

Part C: Accessibility (A) — Pediatric Ergonomics & Offline Equity

Evaluated by parents and educators on a 1–5 scale to verify inclusive design and socioeconomic equity:

- **[ACC-01] Pediatric Touch Targets:** The interactive buttons, letter cards, and microphone icon are large enough ($\geq 64\text{dp}$) for small child fingers to tap comfortably without accidental miss-taps.
- **[ACC-02] Non-Verbal Audio Dual-Coding:** Spoken voice instructions and visual animations allow young, non-reading children to understand tasks without needing to read written on-screen text.
- **[ACC-03] High Contrast & Readability:** High-contrast UI colors and large typography satisfy WCAG 2.1 AA standards, ensuring readability under indoor lighting and outdoor glare.
- **[ACC-04] 100% Offline Access & Equity:** Operating 100% offline without requiring internet, WiFi, or mobile data credits provides equitable, cost-free learning opportunity for low-resource households.
- **[ACC-05] Child Safety & Privacy:** The total absence of third-party advertisements, external web links, and in-app purchases ensures a secure, distraction-free environment for young learners.

Part D: Child Observational Task Rubric & Time-on-Task (Observer Rubric)

The researcher or parent observer records objective behavioral performance across Letter Node 1 (M) and Chapter 1 Blend It:

- **[OBS-01] Level Map Entry:** Child independently taps active glowing node. Stopwatch: Time from app launch to node tap (seconds).
- **[OBS-02] Hear It Audio Engagement:** Child listens to phoneme sound modeling (/m/ for Mouse) and repeats sound. Rated: High, Neutral, Distracted.
- **[OBS-03] Say It Microphone Vocalization:** Child presses mic button and produces verbal phoneme response. Stopwatch: Mic tap to ASR result (seconds).
- **[OBS-04] Find It Picture Discrimination:** Child discriminates target pictures from distractors. Rated: 3/3 on first try, 1-2 mistakes, or needed help.
- **[OBS-05] Blend It CVC Word Assembly:** Child arranges letter tiles into correct CVC word (SAM). Stopwatch: Display to submission (seconds).

7. Validation Testing Protocol & Field Administration Guidelines

- **Step 1: Test Environment & Device Provisioning:** 5 Android test devices (tablets and smartphones running Android 8.0+) are pre-loaded with playit-debug.apk. The testing space is verified to have ambient noise $\leq 40\text{dB}$ using the app's built-in noise threshold indicator.
- **Step 2: Informed Consent & Demographic Intake:** Parents and teachers complete the digital consent form (Q1-Q2) and demographic profile (Q3-Q5) before any testing begins. Child verbal assent is confirmed.
- **Step 3: Diagnostic Pre-Test (2 Minutes):** The observer administers the 10-item baseline assessment (Phoneme Flashcards M, S, A, I, B and Word Cards SAM, BUS) to record pre-intervention reading knowledge.
- **Step 4: Observational Gameplay Session (15 Minutes):** The child completes Letter Node 1 (Hear It $\rightarrow$ Say It $\rightarrow$ Find It) and the Chapter 1 Blend It challenge. The observer records OBS-01 to OBS-05 timings with a stopwatch.
- **Step 5: Diagnostic Post-Test & Affective Feedback:** Immediately post-play, the observer re-tests the phonics items to compute the learning gain delta, followed by the 3-point Smileometer face rating.
- **Step 6: Caregiver SUS & Teacher Survey:** Parents and teachers complete the 10-item SUS questionnaire and accessibility checklist via Google Forms synced to the master Google Sheet.

8. Thesis Chapter 4 Statistical Integration & Refactoring Roadmap

The empirical data gathered from the 30-respondent validation cohort will directly substantiate **Thesis Chapter 4 (Results and Discussion)** and drive the **Week 3 Requirements Refactoring**:

- **Inferential Phonics Gain Analysis:** Pre-test and post-test scores will be evaluated via a **Paired Samples t-test** (or Wilcoxon signed-rank test if normality is violated) to determine whether PlayIT yields statistically significant reading gains ($\alpha = 0.05$).
- **Caregiver Usability Benchmarking:** SUS survey responses will be scored using Brooke's normalized algorithm to generate composite usability ratings, benchmarked against the standard Bangor et al. (2008) curved grading scale (Target: ≥ 75.0 / Grade B+).
- **Software Quality Verification:** Offline Vosk latency (SYS-01 $\leq 0.5\text{s}$), 60 FPS animation stability (SYS-02), and SQLite/Room transactional integrity (SYS-03, 0% data loss) will be verified across test device logs.
- **Week 3 Document Refactoring:** Findings will be synthesized into the **SRS v3.0 Requirements Traceability Matrix (RTM)** (officially incorporating FR-13 Blend It and FR-14 Multi-Profile) and the **SDD v2.0 Architecture Specification** due on September 19, 2026.